

Song Success Prediction Model

Project Proposal

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In this project I will analyze the [Spotify Music Analytics Dataset \(2015–2025\)](#) and combine with country data from the [Global Data API](#) in order to create a prediction model that will help artists, producers, and record labels determine a song's potential success. The Spotify Music Analytics Dataset will provide data for existing song features and their popularity, the Global Data API will provide key data from the different regions that songs are popular.

Problem Statement

Music artists, producers, and record labels must make major decisions in the early stages of song production with limited information and high uncertainty. From selecting final versions to be released, to budgets, to targeted regions, these decisions are typically made through intuitive guesses rather than backed by data driven insights. This project aims to develop a predictive model that evaluates a new song's likelihood of success by estimating its overall popularity and identifying regions where it is most likely to succeed.

Context

The growth of popularity of music streaming over the last decade has generated a massive amount of data in customer listening behavior, engagement, and song preferences across regions. Spotify is one of the most popular music streaming subscription services making the data generated from that source ideal for analysis of song features and how they correlate to popularity. By combining song analytics with country data, it is possible to create a data-driven decision-support tool that helps stakeholders anticipate performance outcomes before or shortly after release. This project is designed for record labels, producers, and artists seeking to optimize promotion strategies and maximize return on investment.

Criteria for Success

- Delivery of a predictive model that can estimate song popularity with reasonable accuracy using pre-release or early-release features
- The model identifies geographic regions where a song is most likely to perform well

Scope of Solution Space

- Data Wrangling
 - Import, clean, and pre-process Spotify Music Analytics Dataset
 - Query Country-level data from the Global Data API and successfully combine with Spotify data
- Data Exploration
 - Discover trends within datasets for song popularity by genre, region, and other useful insights to help artists, producers, and labels make early decisions.
- Modeling
 - Regression and classification modeling for song popularity

Constraints within Solution Space

- Song success is influenced by external factors (marketing spend, artist reputation, social media trends) that may not be fully captured in the data
- Predictions will be probabilistic rather than deterministic

Stakeholders

- Music Artists
- Music Producers
- Record Label Executives

Key Data Sources

- **Spotify Music Analytics Dataset (2015–2025)**
Provides song-level popularity metrics and audio features used as primary predictive inputs.
<https://www.kaggle.com/datasets/rohiteng/spotify-music-analytics-dataset-20152025>
- **API Countries**
Supplies country-level metadata that can be used to enhance regional analysis and contextualize performance predictions.
<https://www.apicountries.com/>